Conjecture 00000000147 Does Not Specify a Probability Model

quantumtool

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The statement is undefined as written

The conjecture writes $\mathbb{P}(g_n > t \log p_n)$ but does not specify a probability space, random variable, or sampling rule. Prime gaps g_n and primes p_n are deterministic. Thus the probability in the displayed expression is not defined by the data of the conjecture.

Even if one assigns the deterministic event its natural point-mass probability for each n, its value is either 0 or 1. For every fixed $t > 0$, however, e^{-t} is strictly between 0 and 1. A sequence of values confined to $\{0, 1\}$ cannot converge to this interior value: each term stays at least $\min(e^{-t}, 1 - e^{-t})$ away from it. A meaningful tail conjecture would need to specify a random model or an empirical distribution over a growing range of indices; these choices define different statements.

Formalization scope

The Lean file checks the arithmetic core of the point-mass interpretation: a deterministic event probability, encoded as 0 or 1, cannot lie in the central interval corresponding to a fixed positive tolerance around 1/2. The absence of a probability space and the interpretation choice are discussed above, not represented as a Lean measure-theory theorem.

Lean formalization

Lean formalizes the separation of Boolean probabilities (0 or 1) from the open interval around 1/2. The problem statement's missing sample space and the conditional conclusion about its intended interpretation are explained in the report.